

2009 H1 Biology Preliminary examination Paper 2 SMS

Question 1 [Total: 12]

(a) Name the structures labeled **A** to **D**. [2]

A RER/ ribosomes

C Golgi body / apparatus

B Mitochondria

D Microvilli

@ ½ m

(b) Describe briefly how the two structures, **A** and **C**, are related in the cell. [3]

Polypeptides synthesized at ribosomes are channeled into RER;

where post-translational modifications/ref. to glycosylation take place;

Intracellular transport via interconnected cisternae ;

and transport vesicles to Golgi apparatus;

where final products (e.g. ref. to lysozyme) are formed and packaged (into secretory vesicles/lysosomes) ; @ 1m

(c) With reference to Fig. 1.1, explain the significance of the numerous **B** found adjacent to **A**. [1]

Synthesis of ATP;

required for protein synthesis/translation at ribosome / RER; @ ½ m

(d) (i) State the function of structure **D**. [2]

Increase surface area to volume ratio of cell surface membrane of the cell;
[1]

Ref. to uptake of material/nutrients via diffusion and active transport;

(e) Name the molecule labelled P. [1]

β-glucose [1], reject glucose

(f) Name the bond labelled Q and outline how it is formed. [1]

**Bond formed: β - (1, 4) glycosidic bond [0.5],
Process: Condensation / polymerisation [0.5]**

(g) Using the information from fig. 1.4, explain why cellulose is suitable as material for the plant cell wall. [2]

Strength – straight chain polymer; multiple hydroxyl groups [0.5] on the glucose residues from one chain form hydrogen bonds [0.5]; with oxygen molecules on the same or on a neighbor[0.5] chain, holding the chains firmly together side-by-side and forming microfibrils, followed by macrofibrils with high tensile strength[0.5].

The overall structure is a mesh of bundles of fibrils. Within this network are extensive pores which allow movement of water and minerals.

Question 2 [Total: 5]

(a) Explain the sequence of events which occur from the G₁ phase till the cell reach the M phase of the cell cycle. [2]

G₁ phase:

- new organelles are being synthesized
- Increase in the synthesis of structural proteins and enzymes needed for DNA replication
- Free deoxyribonucleotides synthesized

S phase:

- DNA replication takes place
- each chromosome now has two identical sister chromatids, DNA content doubled

G₂ phase

- the production of microtubules needed for process of mitosis
- new organelles are being synthesized e.g. centrioles

@ ½ m each

- (b) Starting with the cell closest to the beginning of mitosis, arrange the letters **P, Q, R** and **S** in the order in which they would occur during the cell cycle.

[1]

Q, S, P, R

- (c) Explain how stage **P** ensures that the daughter cells formed will be genetically identical. [2]

Stage P: anaphase

- Each chromosome has 2 genetically identical sister chromatids held together by a centromere.
- Centromere of each chromosome divides (→ each sister chromatid has its own centromere)
- Microtubules / spindle fibres attach to centromere of each sister chromatid to pull the chromatids to each pole
- ensures equal distribution of genetically identical DNA in each new daughter cell.

Question 3 [Total: 10]

- (a) State a phenotypic characteristic of *L. genimaculata* and suggest a possible selection pressure acting on this characteristic. [2]

- **phenotypic characteristic: colour of skin**
- **selection pressure: camouflaged from predators**

- (b) Examine the data in **Fig. 3.1** and describe **two** differences between the mating calls of *L. myola* and *L. genimaculata*. [2]

Two of *L. myola* has:

- **a longer mating call than *L. genimaculata***
- **a faster note rate than *L. genimaculata***
- **a higher pitch than *L. genimaculata*.** **2m**

- (c) Suggest how changing climate conditions could have isolated small populations of rainforest frogs. [1]

Either of:

- Pockets of rainforest may have changed and become more or less favoured by frogs
- Some ponds may have dried up restricting location of frogs.

*The condition that isolated the groups had to be **permanent**, due to **climate change**, and relevant. Some incorrect suggestions were flooding (not a permanent event), mountains, (not climatic) and volcanoes (not relevant).*

- (d) What is one type of evidence that biologists could use to estimate an approximate date of divergence of these two species? [1]

Any 1: Comparing extent of differences in:

- mitochondrial DNA
- DNA
- amino acid sequence.

or

A description of fossil or stratigraphic evidence, such as ‘the sequence of fossils are observed to determine when divergence occurred’.

- (e) Describe how *L. myola* could have evolved from *L. genimaculata*. (Use the terms ‘allele frequency’ and ‘gene flow’ in your answer.) [3]

- For many generations, a small population of *L. genimaculata* was isolated by changed geography due to climate;
- and there was no gene flow (or allele flow) between the populations
- allele frequencies in the small population changed (particularly with respect to size and mating call)
- due to natural selection or genetic drift
- Eventually the two groups are genetically different enough
- to prevent successful reproduction.

@ ½ m each

(f) How could biologists determine this experimentally? [1]

Either of:

- interbreed the two species and if they produce viable and fertile offspring, they are the same species
- interbreed the two species and if they do not produce viable or fertile offspring, they are not the same species.

To gain the mark for this question students needed to state what could be concluded from their suggestion.

Question 4 [Total: 7]

(a) Using the symbols **C^p** for purple butterflies and **C^w** for white butterflies, **R** for round-shaped wings and **r** for heart-shaped wings, state the genotype and phenotype of the **F₁** generation. [2]

F₁ genotype: C^pC^w Rr

F₁ phenotype: Opalescent with round-shaped wings

- (b) Draw a genetic diagram to explain the results from the cross between the F₁ generation. [4]

Parental phenotype	Purple, round – shaped butterfly	x	White, heart – shaped butterfly
Parental genotype	C ^p C ^p RR	x	C ^w C ^w rr
Gametes (circled)	C ^p R		C ^w r
F1 genotype	C ^p C ^w Rr		
F1 cross	Opalescent, round-shaped butterfly C ^p C ^w Rr	x	Opalescent, round-shaped butterfly C ^p C ^w Rr
Gametes (circled) 1m	C ^p r C ^w r C ^p R C ^w R	C ^p r C ^w r C ^p R C ^w R	

Punnett square (F₂ generation):

	C ^p R	C ^p r	C ^w R	C ^w r
C ^p R	C ^p C ^p RR Purple, round shaped	C ^p C ^p Rr Purple, round shaped	C ^p C ^w RR Opalescent, round shaped	C ^p C ^w Rr Opalescent, round shaped
C ^p r	C ^p C ^p Rr Purple, round shaped	C ^p C ^p rr Purple, heart shaped	C ^p C ^w Rr Opalescent, round shaped	C ^p C ^w rr Opalescent, heart shaped
C ^w R	C ^p C ^w RR Opalescent, round shaped	C ^p C ^w Rr Opalescent, round shaped	C ^w C ^w RR White, round shaped	C ^w C ^w RR White, round shaped
C ^w r	C ^p C ^w Rr Opalescent, round shaped	C ^p C ^w rr Opalescent, heart shaped	C ^w C ^w Rr White, round shaped	C ^w C ^w rr White, heart shaped

Correct genotypes in F₂ generation with correct corresponding phenotypes: 1m each

F₂ Phenotypic ratio (1m):

Purple, round shaped	purple heart shaped	opalescent, round shaped	opalescent, heart shaped	white, round shaped	white, heart shaped
3	: 1	: 6	: 2	: 3	: 1

(c) Suggest an explanation for the results observed in the F₂ generation. [1]

- Random fusion of two (of 4) gametes produced six different kinds of phenotypes in the F₂ generation
- Butterflies with the genotype C^PC^W will exhibit co-dominance, thus giving rise to opalescent appearance in terms of color

Question 5 [Total: 6]

(a) Describe two unique features of stem cells that could be seen from the figure above. [2]

- Stem cells can undergo differentiation to give rise to multipotent cell types.;[1]
- Multipotent stem cells cannot revert into embryonic stem cells but embryonic stem cells can form multipotent stem cells. [1]
- Stem cells are capable of dividing indefinitely and producing copies of themselves / capable of self renewal. [1]

(b) Describe the developmental potential of the one-celled zygote and the inner cell mass of the blastocyst. [4]

- One celled zygote will divide by mitosis to give rise to a blastocyst;
- The inner cell mass of the blastocyst are all pluripotent/totipotent embryonic stem cells;
- that will differentiate into all/most cell types in a human being;
- given the right cues/signals such as growth factors or hormones.

(d) Adult stem cells are multipotent stem cells found among differentiated cells in a tissue or organ. What are the primary roles of these adult stem cells and describe one appropriate unique feature of this type of stem cells that is not mentioned in (a). [2]

- Their primary role is to maintain and repair the specific tissues in which the stem cells are found in.
- They can remain undifferentiated / unspecialized for long periods of time.
- Adult stem cells are multipotent stem cells that can divide by mitosis to give differentiate into the tissues or cell types
- Multipotent stem cells cannot revert into embryonic stem cells but embryonic stem cells can form multipotent stem cells. [1]* accepted if it is not mentioned in part (a).

Question 6

(a) Describe the role of NAD and NADP in cells. [6]

- o NAD is a **co-enzyme**
- o **NAD are electron carriers which provided the reducing powers for the reactions.**
- o **form intermediates in glycolysis, linked reaction and Krebs cycle aerobic respiration)**
- o **NAD carries electrons to the Electron Transport Chain for oxidative phosphorylation**
- o **Role of NAD in Glycolysis as oxidizing agent to convert GALP to G3P.**
- o **Role of NAD in Linked reaction to convert pyruvate to acetyl coA.**
- o **Role of NAD in Krebs cycle as oxidizing agent to oxidize various intermediates.**
- o **Role of oxidative phosphorylation, where NADH and e⁻ to electron carriers for the synthesis of ATP.**
- o **For the synthesis of 30 out of the 38 ATP produced during the complete oxidation of 1 glucose molecule from 10 molecules of NADH produced in glycolysis, link reaction & krebs cycle**
- o **Reoxidized NAD must return to glycolysis, to keep the processes of anaerobic respiration going**

[Max 3]

- o **NADP is a coenzyme**
- o **that serves as an electron acceptor in the electron transport chain during photophosphorylation**
- o **Electron is transferred to a primary electron acceptor which in turn donates it to a series of electron carriers.**
- o **The electron carriers are of progressively lower energy levels.**
- o **During photolysis of water, the protons from the splitting of the water molecule formed reduced NADP (or NADPH) along with the electrons from PSI that combined with the NADP.**
- o **The reduced NADP then passes along the high energy electrons on to the Calvin cycle for the reduction of glycerate-3-phosphate (GP) to another 3C sugar called triose phosphate (TP).**
- o **NADP then returns to the light dependent stage to pick up another hydrogen ions.**

(b) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [8]

- o Definition of limiting factor, the rate of reaction will increase when the concentration of that factor increases.
- o and when the rate of photosynthesis is limited by more than one factors
- o the rate is limited by the factor which is in the shortest supply

Light intensity

- o directly affects light-dependent stage of photosynthesis
- o at low light intensities, light intensity is the limiting factor
- o rate of photosynthesis increases proportionally with increasing light intensity
- o at high light intensities, rate of photosynthesis decreases
- o further increase in light intensity results in the light saturation point being reached
- o beyond this point, further increase in light intensity will have no effect on the rate of photosynthesis
- o light intensity is no longer a limiting factor
- o other factors will become the limiting factor

carbon dioxide concentration

- o a major limiting factor since atmospheric CO₂ concentration is low
- o CO₂ is needed in the light-dependent stage / Calvin cycle
- o it is reduced to carbohydrate / sugar
- o it is fixed by combining with ribulose-1,5-bisphosphate (RuBP)
- o increase in CO₂ concentration results in a proportional increase in the rate of photosynthesis
- o when the carbon dioxide concentration is high and the Calvin cycle is already performing at the optimal level,
- o further increase in carbon dioxide concentration will have no effect on the rate of photosynthesis
- o it is no longer a limiting factor
- o other factors will become the limiting factor

(c) Explain the small yield of ATP from anaerobic respiration. [6]

- o Glucose is **not completely oxidised**, hence a lot of energy is not released but **stored in intermediate molecules** like **lactate in mammals, ethanol in yeast**.
- o **In absence of oxygen, pyruvate** is unable to enter mitochondria **and cannot undergo the linked reaction, Krebs Cycle and oxidative phosphorylation.**
- o **The pyruvate is channelled to the fermentation to regenerate NAD and is partially broken down** into lactate in animals or alcohol in plants.
- o **ATP in glycolysis** comes from **only two steps, by substrate level phosphorylation.**
- o **1, 3 – diphosphoglycerate (3C) is converted to 3– phosphoglycerate (3C).**
- o **3– phosphoglycerate (3C) is converted to pyruvate.**
- o **There is a loss of two ATP right at the start of glycolysis in the priming reaction, hence net yield is only $4-2 = 2$ ATP.**

Question 7

(a) Describe the eukaryotic processing of pre-mRNA [6m]

1. occurs in nucleus of eukaryotic cell;

Capping

2. 5' end of pre-mRNA transcript is capped by addition of a guanylyl residue;

3. cap structure is methylated at the 7-position of the G residue;

4. forming 5' end of pre-mRNA with methylated guanine cap;

Polyadenylation

5. adding of poly-A tail to 3' end of pre-mRNA;

6. poly A addition signal is at the consensus sequence of AAUAAA upstream of addition site;

7. prevent enzymatic degradation in cytoplasm;

RNA Splicing

8. removing of introns (non-coding regions) of pre-mRNA;

9. by spliceosomes at specific splice sites;

10. splicing/joining of exons (coding regions) of pre-mRNA;

Alternative splicing

11. takes place to include different exons from a pre-mRNA;

12. increases coding capacity of genome.

13. allows variations of proteins to be produced, increasing protein diversity;

14. mature mRNA formed leaving nucleus through nuclear pores into cytoplasm;

(b) Describe the control mechanisms involved in the expression of eukaryotic genes. [8]

Chromatin modification	- Plays a significant role in transcriptional regulation. - Genes in the heterochromatin region, which is more highly condensed, are usually not expressed - DNA methylation prevents gene expression when methyl groups are added to the DNA - DNA methylation is essential for the long term inactivation of genes - Histone acetylation increases the expression by adding acetyl groups to histones - which affect how compacted the region is, therefore affect transcriptional activation
Regulatory sequences	2 types of control elements present: - Basal / general transcription factors bind to proximal control elements which are required for transcription of all genes - Specific transcription factors (activators) bind to distal control elements (enhancer) which act in a tissue-specific / time-specific manner. These are protein activators which bind to distal control elements associated with specific genes, and activate transcription of these specific genes. - This aids in the folding of DNA which aids in RNA polymerase accessibility
Induction / repression in response to external stimuli	In response to external stimuli, transcription factors may bind to (or silencer) regions, thereby activating (or silencing) transcription
Post transcription	Alternative RNA splicing: Different mRNA molecules are produced from the same primary transcript Regulation of mRNA degradation - enzymatic shortening of the poly (A) tail which triggers the action of enzymes to remove the 5' cap - Once the cap is removed, nuclease enzymes chew up the mRNA from the 5' end

Translation	Control binding of ribosome/prevent ribosome from translating/block ribosome attachment/ initiator tRNA attachment by regulatory proteins at the leader regions at 5' end of the mRNA
Post translation	Activity of protein can be regulated by (a) end product inhibition (b) covalent modification /phosphorylation (c) Protein tagged with ubiquitin and recognised by proteasome will be degraded: regulates length of time protein functions in the cell.

Max 8m: to obtain full marks, student should have a good discussion of aspects of eukaryotic gene expression at all levels

(c) How does prokaryotic DNA differ from eukaryotic chromosome? [6]

	Features	Prokaryotes	Eukaryotes
	DNA organisation	Prokaryotic DNA is organized into a DNA-protein complex called the nucleoid. The 'chromosome'(DNA) is undergoes supercoiling and is complexed with DNA binding proteins.	Eukaryotic chromosomes consist of a DNA that is coiled around groups of 8 histones (nucleosomes) and further coiled around proteins into chromosomes.
	<u>Genome size not just size which may mean size of the organism</u>	Prokaryotic genomes are much smaller, 0.6 to 10Mb.	Smallest eukaryotic genomes being less than 10 Mb in length, and the largest over 100 000 Mb
	<u>Gene length not DNA length</u>	In general, prokaryotic genes are shorter than their eukaryotic counterparts, the average length of a bacterial gene being about two-thirds that of a eukaryotic gene, even after the introns have been removed from the latter.	Longer gene sequences compared to prokaryotes
	Introns	Coding sequence proceeds from start to finish without interruption by introns.	Some of the non-coding DNA in eukaryotes is present as introns within genes.
	Repetitive sequences	Infrequency of repetitive sequences.	Repetitive DNA make up 59% of the genome in humans
	DNA codes	Most of DNA codes for protein, tRNA, or rRNA. Small amount of non-coding DNA consists mainly of regulatory sequences, such as promoters	Most of DNA does not code for protein or RNA, and it includes more complex regulatory sequences. Humans have 10000 times as much non-coding DNA as prokaryotes.
	Nucleus	Absence of membrane-bound nucleus, DNA localized in zone of cytoplasm called nucleoid	Presence of membrane-bound nucleus containing chromosomes
	Supportive structure	DNA is held in place by association with DNA-binding proteins	DNA is associated with proteins called histones and held together via ionic bonds
	Origins of replication	One	Many